

Elioth Krahler

Bismarck, ND 58504 (Willing to Relocate) // eliothkrahler@gmail.com // krahler.com // github.com/EliothK // linkedin.com/in/elioth-krahler-a9215843a

Summary

Computer Science graduate who learns whatever the problem needs, then ships it and writes up what broke. Built and run krahler.com end to end: a Spring Boot API and React site on Azure, provisioned with Terraform and deployed through GitHub Actions with staging, approval-gated infrastructure changes and passwordless database access. Also builds and evaluates ML systems in Python, PyTorch and XGBoost, from solar forecasting to a physics-informed reactor surrogate, always against a baseline. Looking for ML engineering, backend, fullstack or DevOps work.

Education & Certifications

Bachelor of Science in Computer Science // Western Governors University	2022 – 2026
WGU Certificate : AI Optimization Developer	2026
WGU Certificate : Java Developer	2026
Peoplecert : ITIL 4 Foundations	2026
Linux Professional Institute : Linux Essentials	2025

Technical Skills

Languages

- Java, Python, TypeScript, C#, SQL, HTML, CSS

Frameworks/Libraries

- Spring Boot, Spring Data JPA, Flyway, JUnit 5, React, Vitest, PyTorch, TensorFlow, XGBoost, Prophet, scikit-learn, NumPy

Tools & Infrastructure

- Terraform, GitHub Actions (CI/CD, OIDC), Microsoft Azure (Static Web Apps, Container Apps, Azure SQL, Entra ID), Kubernetes/AKS, Docker, Podman, Git/GitHub/GitLab, Apache Maven, Linux, NVIDIA GPU Acceleration, Windows Networking, Visual Studio Code, AI Coding Assistants (Claude, GitHub Copilot, Cursor)

Core Competencies

- AI/ML Optimization, AI Data interpretation, Neural Network Architecture & Deep Learning, API Integration, Machine Learning Algorithms, Predictive Modeling
- Data Structures & Algorithms, Software Design & Architecture, Object-Oriented Design & Design Patterns, Computer Architecture, Operating Systems Concepts (Process-Memory Management, Concurrency)
- Relational Database Design & Normalization, Unit Testing, Multi-threading, Data Wrangling, Data Cleaning, Model Refinement, Algorithmic Optimization
- IT Governance & Business-IT Alignment, Troubleshooting, Hardware Maintenance

Technical Projects

Krahler.com // Cloud Platform & DevOps Pipeline // GitHub // Solo Developer

- Built and operate krahler.com: a prerendered React/TypeScript site on Azure Static Web Apps and a Spring Boot 4 (Java 25) REST API on scale-to-zero Azure Container Apps, backed by Azure SQL serverless with Flyway migrations.
- Provisioned all infrastructure with Terraform in separate state roots; every pull request posts a read-only plan, and changes apply only after manual approval and a check that the plan hasn't changed since it was approved.
- Built GitHub Actions CI/CD with OIDC and no stored cloud credentials: lint and unit tests (Vitest, JUnit 5), a staging environment for every pull request, and post-deploy smoke, real-browser and Lighthouse gates (mobile performance 0.88–0.90, accessibility 1.00).
- Hardened the API: passwordless database sign-in through a managed identity, fixes for an email header injection and a rate-limit bypass found in my own security review, and a cost fix for a scheduled job that kept the serverless database from ever pausing.
- First ran it on Azure Kubernetes Service (Gateway API, cert-manager, Trivy image scanning, digest-pinned rollouts), then moved it to static hosting when the cost stopped making sense; the AKS setup is kept as an on-demand Terraform lab.

NeutroSurrogate // Physics-Informed ML Surrogate // GitHub // Lead Developer

- Wrote a finite-volume neutron diffusion solver from scratch in Python and verified it against analytic theory (under 1 pcm, second-order convergence) before training any model on it.
- Trained a PyTorch network, tuned with cross-validated Optuna searches, that predicts k-eff and the flux shape of 16-zone reactor designs: a 274 pcm median error on 3,000 unseen designs, 11x better than gradient-boosted trees and 17x better than linear regression.
- Added a Rayleigh-Ritz correction that turns the predicted flux into k-eff within 1.5 to 2.3 pcm RMSE (about 0.002%) at 25x the solver's speed; tested in CI with pytest.

SolarCast // AI Forecasting // GitHub // Portfolio Project

- Built a portfolio forecasting system that corrects the free Open-Meteo weather forecast against NREL NSRDB satellite measurements, predicting solar irradiance 6 to 48 hours ahead with an XGBoost and LSTM ensemble.
- Rebuilt six notebooks as a tested Python package with CI: one command downloads the data, tunes (Optuna, Keras Tuner), trains and evaluates any site from its latitude and longitude.
- Trained and ran the full pipeline on my own Linux machine with an NVIDIA GPU, using only public data feeds and no cloud services.
- Cut RMSE 24% below the Open-Meteo forecast alone, 6 and 12 hours ahead, on a year of unseen data (83 vs 110 W/m² in daylight); reported that long-range models don't beat a seasonal average.

Professional/Work Experience

Desktop Refresh Technician // Resource Revolution // Minneapolis, MN 2021 – 2022

- Expanded role to include leadership of the Refresh team and served as a backup for IT during staff shortages.
- Developed PowerShell commands to automate and expedite the desktop refresh process, improving team efficiency.

Delivery Driver // DoorDash // Saint Paul, MN 2022 – Present

- Independent delivery driver. Kept it going alongside a full CS degree, and now alongside my job search and the projects on krahler.com.